

Selection on Exit and Sticky Prices

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Product turnover and the real effects of nominal shocks

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Does the composition of exit (and entry) change the real effects of an aggregate nominal shock?

This paper

Facts: Dutch scanner data, 2008–2020, 163M transactions, about 1M products.

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Mechanism: turnover replaces prices too.

Random turnover reprices products that would have kept a wrong price.

Selective turnover reprices products that were about to fix it anyway.

Main results

Amplification: output responds **15%** more to a nominal shock than under **random** turnover.

One-year cumulative response of output to a permanent 1% shock to nominal spending.

Same exit rate, same pricing moments as the **random** economy: only composition differs.

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State- vs time-dependent pricing: under Calvo the amplification survives but shrinks to **2%**.

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Takeaway: money is less neutral than models without selection say.

Policies that blunt selection weaken monetary policy.

Especially when inflation is low.

Substantive: new source of amplification, **exit composition** is a channel of monetary transmission.

Amplification via the pricing problem: Golosov and Lucas (2007); Nakamura and Steinsson (2010); Mongey (2021); Wang and Werning (2022); Aruoba et al. (2024); Blanco et al. (2024, 2025).

Amount of turnover: Read (2020); Bilbiie (2021); Khalil and Lewis (2024).

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Technical: firm dynamics meets nominal rigidities.

1. First menu cost economy with selective entry and exit and aggregate risk.

Selection on exit (Hopenhayn, 1992; Clementi and Palazzo, 2016) and entry (Sedláček and Sterk, 2017; Sterk et al., 2021). Selection measured in productivity units: Caballero and Hammour (1994); Foster et al. (2016); Hamano and Zanetti (2022); Hartwig and Lieberknecht (2025).

Much less work on what selection does to monetary non-neutrality

2. Same selection, different pricing friction: the aggregate answer changes.

State- and time-dependent pricing similar for small shocks, different for large ones: Alvarez et al. (2017); Auclert et al. (2024). Evidence: Cavallo et al. (2024); Gautier et al. (2026)

No work comparing selection across pricing frictions.

Roadmap

- 1 Introduction
- 2 Empirical evidence
- 3 Model
- 4 Low and high inflation regimes
- 5 Conclusion

- One product characteristics predicts both selection and pricing behavior:
 - Product revenues.
- Dutch retail scanner data, 2008–2020 (AiMark): fast-moving consumer goods.
 - 163.5 million non-promotion transactions, ~1M UPCs, 719 categories, 231 chains.
 - About €1.8bn of sector revenue in 2008, out of €30bn.
 - UPC × chain × province; monthly posted price.
- Monthly exit rate 2.0%, monthly entry rate 2.3%, price-change frequency 9.0%.
- Product exit: a year-long losing revenue streak, and national (82%, revenue-weighted).

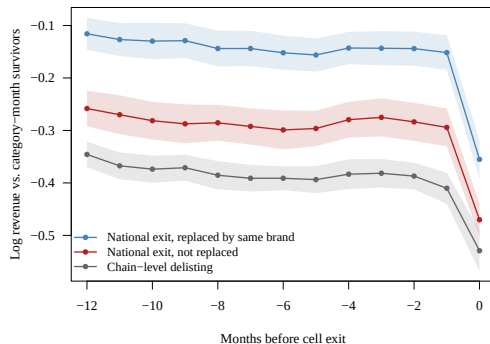
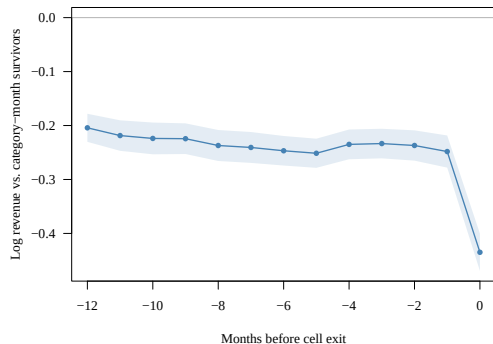
► Filters

► Descriptives

► Whose prices?

► Exit types

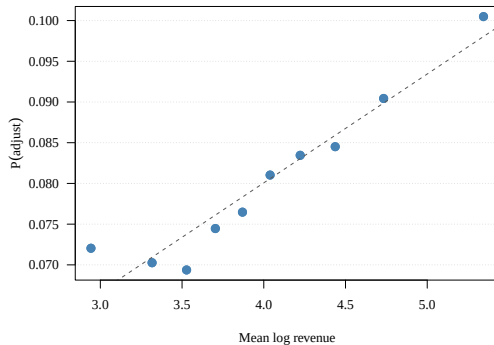
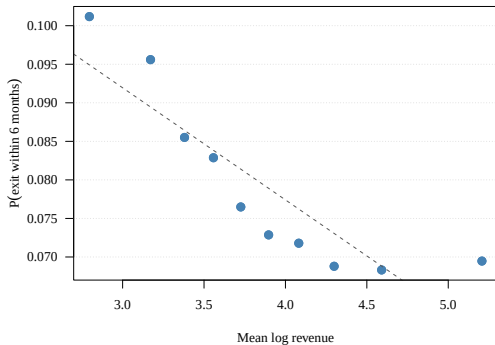
Revenue before exit



Minus 0.2 means selling about 20 percent less.

- Event study on log revenue per UPC-chain-province cell, within category and month.
- Even exits with a same-brand successor sit 12 log points below peers all year.

Low-revenue products exit more and adjust less



► Size of a price change

► Robustness

Taking stock of the data

- Product turnover is large and selective.
- Small products leave more often and reprice less often.
- Exit is anticipated: a product sells below its peers for a year before it goes.
- Even exits with a same-brand replacement sell below their peers.

How much does the composition of exit change the real effects of a nominal shock?

Household and nominal demand

- Representative household: $\max \mathbb{E}_0 \sum_t \beta^t [\log C_t - \chi h_t]$, complete markets, owns the firms.
 - Labor supply $W_t/P_t = \chi C_t$; discount factor $\Xi_t = \beta^t \mathbb{E}_t(C_t/C_{t+1})$.

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- Monetary policy targets nominal spending ($S_t \equiv P_t C_t$)

$$\log S_t = \bar{\pi} + \log S_{t-1} + \sigma_S \eta_t, \quad \eta_t \sim N(0, 1)$$

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- Final good: CES over an endogenous measure μ_t of varieties, $y_{it} = Y_t (p_{it}/P_t)^{-\theta}$.

Nominal GDP level target at a constant nominal rate: the Euler equation gives $1 + i_t = (1/\beta) / \mathbb{E}_t[S_t/S_{t+1}]$, constant under the random walk; optimal under nominal contracting frictions in Bullard and Di Cicio (2025).

Intermediate producers

- Linear labor technology $y = zl$, idiosyncratic AR(1) productivity $\log z' = \rho \log z + \sigma_\varepsilon \varepsilon$.
- Menu cost f in labor units on any price change: $V(s) = \max\{V_N(s), V_A(s)\}$.
- State-dependent exit hazard $\lambda(s)$; pricing internalizes survival through $1 - \lambda(s')$.

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- State-dependent exit hazard $\lambda(s)$; pricing internalizes survival through $1 - \lambda(s')$.
- Entry replaces exit one-for-one in two steps within the period:
 1. Draw $z_1 \sim G_e$, biased toward low productivity: $G_e(z) \propto G(z) \exp(-\tau \log z)$, $\tau \geq 0$.
 2. Price freely at $p_1 = \arg \max_p V_A$, then draw z_2 : entrants start slightly mispriced.

$s = (p_{-1}, z, \mathcal{S})$: inherited price, productivity, aggregate state $\mathcal{S} = (P, Y, W, \mu)$.

Flow profit $\pi(p, z, \mathcal{S}) = \frac{p}{P}y - \frac{W}{P}l - \frac{W}{P}f \mathbb{1}\{p \neq p_{-1}\}$.

$\tau = 0$ entrants draw from ergodic distribution, $\tau > 0$ makes them smaller, as in the data.

► Value functions

Exit hazard

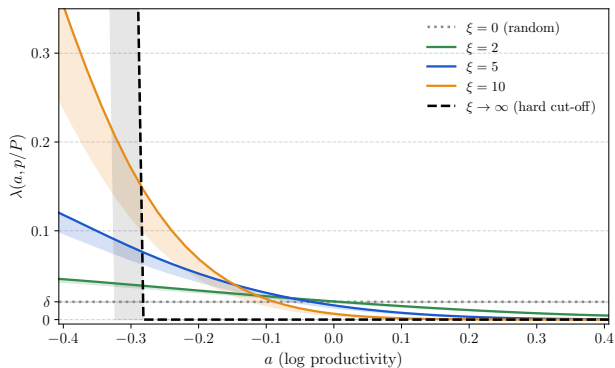
- The hazard is an exponential tilt of the value distribution, normalized to flow δ :

$$\lambda(s) = \delta \cdot \frac{\exp(-\xi V(s))}{\mathbb{E}_\mu[\exp(-\xi V)]}, \quad \mathbb{E}_\mu[\lambda] = \delta$$

- Two parameters: δ fixes the total exit rate, $\xi \geq 0$ the elasticity of selection.
- Same δ , different ξ : same total exit, the economies differ only in who leaves.

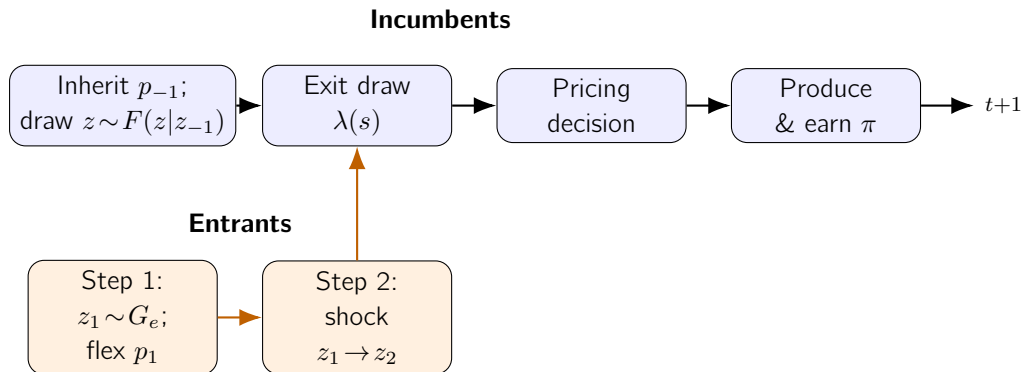
Two limits: $\xi = 0$ is iid random exit at rate δ ; $\xi \rightarrow \infty$ concentrates the hazard on the lowest- V firms, the cutoff rule of Hopenhayn (1992).

Exit hazard by firm productivity



- Band: hazard range inside the inaction region; not adjusting can lower exit risk.

Timing



$$\mu'(s') = \int (1 - \lambda(s)) Q(s, s') d\mu(s) + M \nu_e(s'), \quad M = \int \lambda d\mu$$

Q : transition from pricing policy, productivity draw, aggregate state. ν_e : entrant distribution.

Solving the model

- Heterogeneity and aggregate risk.
 - Tomorrow's price level depends on the whole cross-section (an infinite-dimensional state).
- Krusell and Smith (1998): firms summarize μ by one number and forecast inflation from it.

$$S = S/P_{-1}, \quad \pi = P/P_{-1} = G(S)$$

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1. Given G , solve the Bellman and the stationary μ ;
2. Compute the inflation implied by the CES index at each $\mathcal{S}$;
3. Project that implied inflation on $\mathcal{S}$: the fitted line is the new G ;
4. Iterate until perceived and implied inflation agree.

Definition

A stationary recursive competitive equilibrium consists of values V_N, V_A , a price policy $p(s)$, an exit hazard $\lambda(s)$, an entrant distribution ν_e and mass M , an invariant measure μ , aggregates $\{C, W/P\}$, and a perceived law of motion G , such that

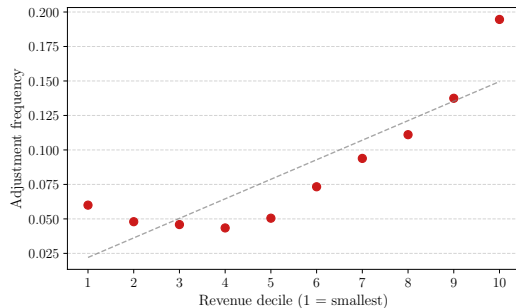
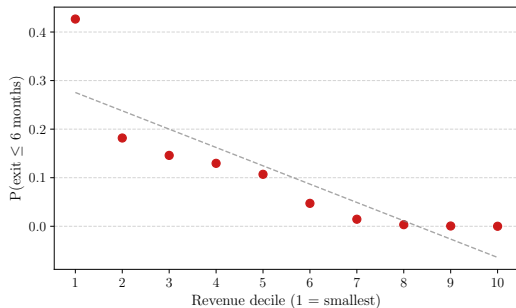
- 1 $C = S/P$ and $W/P = \chi C$ solve the household problem;
- 2 given G , W/P , and λ , the policy $p(s)$ solves the incumbent Bellman;
- 3 ν_e follows the entry block and $M = \int \lambda d\mu$ (one-for-one replacement);
- 4 μ is invariant under the law of motion;
- 5 goods and labor markets clear;
- 6 G is consistent with the inflation generated by the firm policies.

Calibration

Parameter	Description	Value	Target	Data	Model
f	Menu cost	0.155	Frequency of price changes	0.09	0.09
δ	Average exit rate	0.020	Monthly exit rate	0.02	0.02
ξ	Exit selectivity	20	Rel. size of exiting firms	0.66	0.64
τ	Entry selectivity	5	Rel. size of entrants	0.67	0.69
σ_ε	St. dev. productivity	0.086	Mean $\Delta \log p$	0.04	0.04
<i>Untargeted moments</i>					
—	—	—	Mean $ \Delta \log p $	0.10	0.21
—	—	—	St. dev. $\Delta \log p$	0.11	0.21
—	—	—	Fraction upward changes	0.63	0.62
<i>Externally calibrated</i>					
$\bar{\pi}$	Trend nominal demand	0.0028	Nakamura and Steinsson (2010)		
σ_S	St. dev. nominal demand	0.0065	Nakamura and Steinsson (2010)		
ρ	Productivity autocorrelation	0.83	Estimated		
β	Discount factor	0.96	Externally calibrated		
θ	Demand elasticity	4	Externally calibrated		
χ	Labor disutility	1	Externally calibrated		

The empirical facts, untargeted

- The calibration targets only aggregates and relative sizes, distributions are untargeted:



Selection amplifies the response to a nominal shock

- Benchmark ($\xi = 20$, $\tau = 5$) vs. No selection ($\xi = 0$, $\tau = 0$), both at $\delta = 0.02$.
- CIR: one-year cumulated output response to a one-time shock to nominal spending.

Model	CIR (1 year)	vs. No selection
Benchmark	0.178	+14.6%
No selection	0.155	—
No churn	0.156	+0.5%

- Same total exit, same menu cost and pricing moments: +14.6% from composition.
- Long stochastic simulation: the variance of output is 15% higher under selection.

Solved along a deterministic perfect-foresight transition that tracks $\{V_t, \mu_t, \lambda_t, \pi_t\}$; the response is differenced from the contemporaneous no-shock path.

► CIR

► IRF backward-forward shooting algorithm

► $\text{Var}(Y)$

Why selection amplifies

- The hazard enters twice: inside the value function, and inside the law of motion.

Why selection amplifies

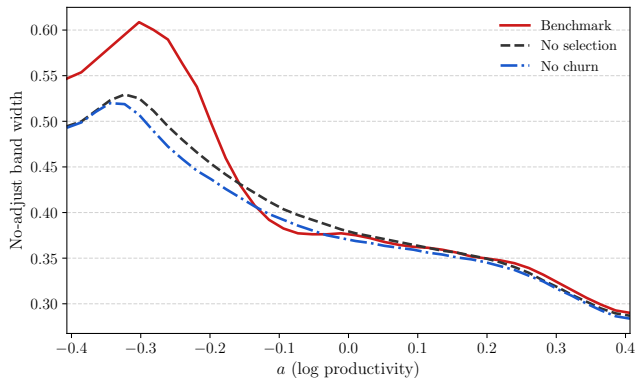
- The hazard enters twice: inside the value function, and inside the law of motion.
 - Composition: every exit hands its slot to an entrant who prices on arrival.
 - Random exit takes the average firm (with a wrong price) and replaces it with a new entrant.
 - Selective exit takes a firm about to reprice: the entrant posts the same correction.
- Same flow δ , so *where* exit lands has different aggregate implications.

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- Anticipation: a high hazard shortens the horizon, so a fresh price is worth less and bands widen.

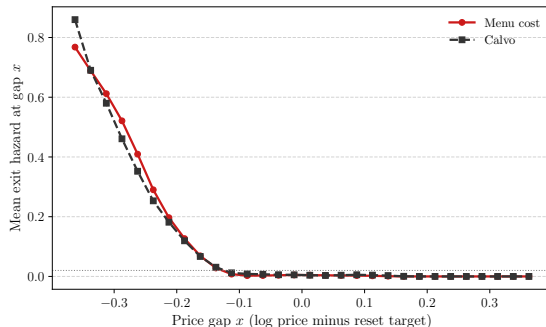
Selection lands on firms just hit by a low productivity draw: marginal cost spikes, they are left underpriced, and they die before they can raise the price.

Anticipation channel: exit risk widens inaction bands



- A high hazard shortens the horizon: paying the menu cost is worth less at low productivity.

Censoring channel: selective exit lands on firms about to adjust



- The measured hazard is negligible over the interior and steep where firms are underpriced.
- Random exit spreads δ evenly, so it mostly deletes interior, persistent mispricing.

► Missing mass

► Who adjusts

► Phantom cohort

Non-neutrality is the standing stock of wrong prices

N.B: CIR $\approx$ amount of mispricing in the economy.

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- I strip the model to the price gap x only.
 - $dx = -\bar{\pi} dt + \sigma dW$, reset to 0 at the band edges $\pm a$, entrants born at 0.
 - Exit at hazard $\lambda(x)$, with the flow held at $\mathbb{E}_\mu[\lambda] = \delta$.

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$$\text{CIR} = \frac{\text{Var}_\mu(x)}{\sigma^2}$$

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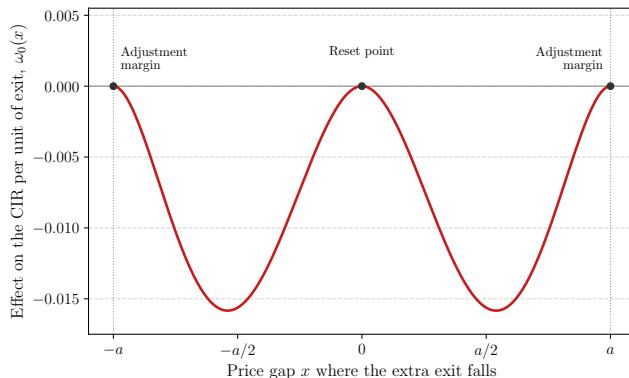
$$\text{CIR} = \frac{\text{Var}_\mu(x)}{\sigma^2}$$

- The response falls in a uniform exit rate, so random churn supplies flexibility:

$$\text{CIR}(\lambda_u) = \frac{a^2}{6\sigma^2} - \frac{a^4}{60\sigma^4} \lambda_u + O(\lambda_u^2)$$

x : posted price minus the free-adjustment price. μ : stationary gap distribution. λ_u : uniform hazard. σ : volatility of gap innovations.

The price of exit at each point of the cross-section



- Zero at the reset point: the firm it kills already carries the entrant's price.
- Zero at the margins: the correction was imminent, so exit relabels it as turnover.
- Negative in between, deepest where the price is wrong and stays wrong the longest.

Whether selection amplifies is the sign of one integral

- Weight the hazard's departure from uniform by the price of exit at each gap:

$$\int_{-a}^a \omega_0(x) (\lambda_s(x) - \lambda_u(x)) dx > 0$$

Theorem (selection amplifies)

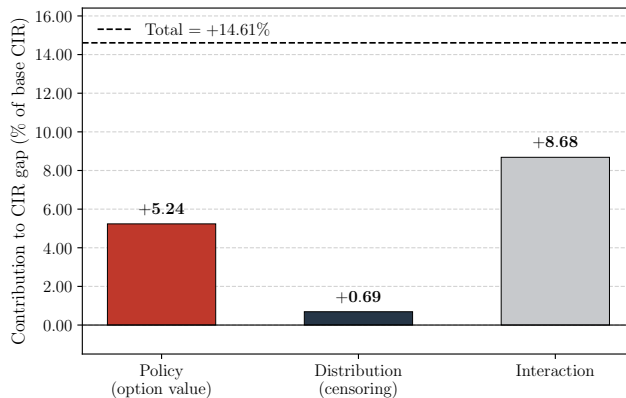
At an equal exit flow and small intensity, if selection concentrates exit near the band edges and spares the interior, then $\text{CIR}_s > \text{CIR}_u$.

It fails when the extra exit sits at intermediate gaps instead.

► Sufficient condition

► Where it breaks

CIR decomposition: complementarity between the channels



- Counterfactuals: firms price under a perceived hazard, mass exits under a realized one.

► Counterfactuals

► Moment decomposition

► Entry vs. exit

Complementarity between the channels

- Selection does two things: prices stay wrong longer, and exit hits firms about to correct.
- Option value alone: prices stay wrong longer, but random exit still clears them.
- Censoring alone: exit only removes prices about to be fixed, but few stay wrong anyway.
- Both: the option value builds the stock that censoring needs.

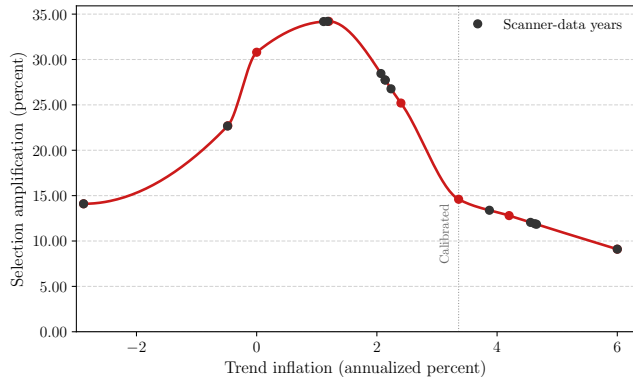
The strongest effect comes from the interaction.

Trend inflation decides how special the exiting firm is

- Amplification needs the firms removed by **selection** to be closer to repricing than the average firm.
- At low inflation repricing is idiosyncratic, so the average firm sits far from its margin.
- High trend inflation pushes every price toward the margin, so the average firm is closer too.
- Experiment: vary trend inflation, hold δ and every other parameter at its calibrated value.

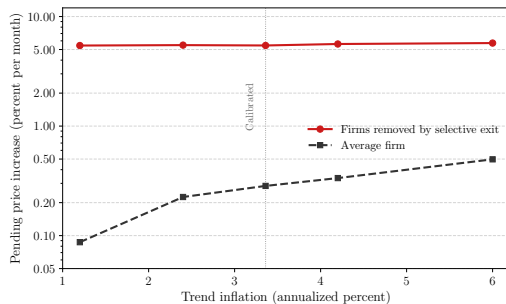
Transmission is strongest at low inflation for a composition reason.

Amplification is largest at low inflation



Trend inflation is annualized. The dots place each sample year at its realized annual inflation level.

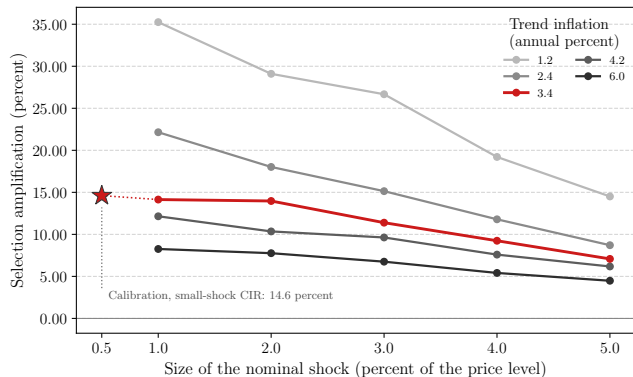
Why the amplification dies at high inflation



- Selection picks the same firm at every trend.
- For higher values of inflation, the average firm has a higher pending price increase.

Each line is the log price increase a firm posts if it survives to the pricing stage.

Large shocks work like high trend inflation



- Larger shocks push all firms to reprice.
- Less room for idiosyncratic reasons to change price: the two exit channels look similar.

Calvo: selection still amplifies

- Replace the menu cost with a Calvo lottery at $\alpha = 0.088$; every other block unchanged.
- Question: how much of the amplification comes from state dependence itself?
- One-year CIR: a 2.4% amplification, against 15% under the menu cost.
- Composition matters under both technologies.

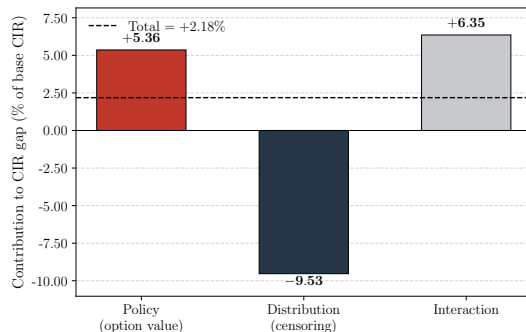
α matches the monthly adjustment frequency of the calibrated menu-cost model and is held fixed across the two Calvo economies, so the nominal friction is constant within the comparison.

► Model

► Moments

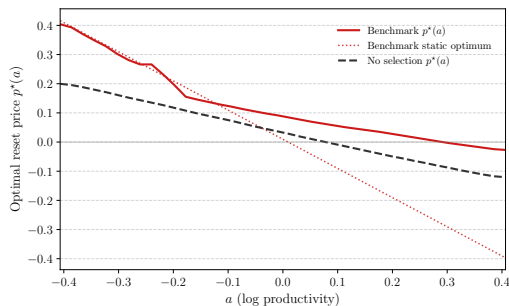
► Persistence

Under Calvo the censoring channel flips sign



- Selection lands on the most misaligned prices at a given productivity.
- A positive shock lowers value even further for firms with larger price gaps.
- Each exit then replaces a worse stale price than without the shock: extra catch-up, a lower CIR.
- Selective exit reduces the overall stock of mispricing compared to random.

The option value moves to the reset price



- Calvo firms cannot move the timing of adjustment, so exit risk lands on the level instead.
- A firm facing near-certain exit expects to die before productivity recovers: it prices statically.
- Under the menu cost the same force widens the inaction band.

Reset price p^* against log productivity; the dotted line is the static flex-price optimum, with slope -1 . The level shift between the curves is general equilibrium: selective exit raises average productivity and the real wage.

Conclusion

- Turnover works a repricing technology.
 - Same exit rate, same pricing moments: 15% larger real effects, 35% at low inflation.
- Monetary expansions concentrate exit on firms stuck with low prices.
- Transmission is strongest at low inflation for a composition reason.
- Policies that weaken selection water down monetary policy.
- The cost of disinflation depends on who exits.

Research agenda

- Firm dynamics and price rigidities

This paper

Markups and Price Discovery (with E. Casanovas)

- Information and beliefs in price setting

Corrective Biases: Overconfidence and the Efficient Use of Information

Mispricing Through Misconfidence (with S. Suetens and B. Uras)

- AI and the market for information

Labor Market Signals (with K. Abbas Nejad, G. Musillo, T. Wicker)

Journal of Labor Economics

AI at the Gate: Platform Answers and the Supply of News (with G. Musillo, T. Wicker)

Moments that Matter: AI and Tail Risk in Portfolio Choice (with H. van Buggenum, B. Uras)

Thank you!



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Appendix slides

When selection amplifies

Adding exit in a direction h , $\lambda \rightarrow \lambda + \theta h$, moves the CIR at rate

$$\left. \frac{d \text{CIR}}{d\theta} \right|_{\theta=0} = \int_{-a}^a \omega(x) h(x) dx, \quad \omega(x) = \frac{f(x)}{\sigma^2} \mathbb{E}_x \left[\int_0^\tau (\text{Var}_f - (x_t - \bar{x})^2) dt \right]$$

At the no-exit baseline it is closed form, $\omega_0(x) = -x^2(a - |x|)^2(a + |x|)/(6\sigma^4 a^2) \leq 0$.

Condition M (margin-concentrated exit)

Relative to the uniform hazard with the same exit flow, $\int_{-a}^a \omega_0(x) (\lambda_s(x) - \lambda_u(x)) dx > 0$.

With $\lambda_s = \theta h$, $\lambda_u = \theta c$ and $c = \mathbb{E}_{f_0}[h]$, Condition M signs the difference at small θ :

$$\text{CIR}_s - \text{CIR}_u = \theta \int_{-a}^a \omega_0(x) (h(x) - c) dx + O(\theta^2)$$

- Any hazard with that profile works: no functional form or other conditions required.
- It compares only placements, not levels: scaling both hazards leaves the sign unchanged.

f : stationary gap density. τ : the next reset, by adjustment or exit.

The same statement without the small-flow restriction

Let $\lambda_\theta = (1 - \theta)\lambda_u + \theta\lambda_s$ and let ω_θ be the kernel at that baseline. Then

$$\text{CIR}_s - \text{CIR}_u = \int_0^1 \int_{-a}^a \omega_\theta (\lambda_s - \delta) dx d\theta$$

- Exact at any exit flow; the local form replaces ω_θ by its value at zero exit.

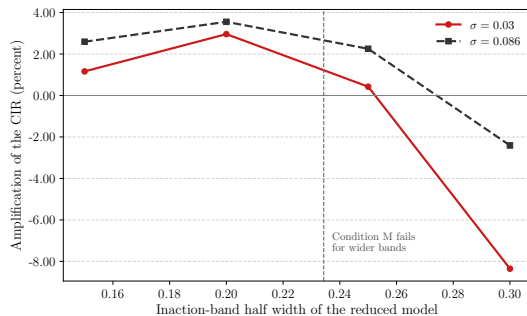
Confine the excess exit to a strip M_ε of width ε at the edges, so $\lambda_s \leq \delta$ on the interior I_ε :

$$\text{CIR}_s - \text{CIR}_u \geq G - \Sigma - K\varepsilon^2 X, \quad G > \Sigma + K\varepsilon^2 X \implies \text{CIR}_s > \text{CIR}_u$$

- G is the exit selection withdraws from where the kernel bites; Σ and X are what it costs.

$G = \int_0^1 \int_{I_\varepsilon} (\omega_\theta)_- (\delta - \lambda_s)$, $\Sigma = \int_0^1 \int_{I_\varepsilon} (\omega_\theta)_+ (\delta - \lambda_s)$, $X = \int_{M_\varepsilon} |\lambda_s - \delta|$, with K depending only on $(\lambda_s, \delta, a, \sigma)$.

The theorem on the measured hazard, and its boundary



- Feeding the measured hazard into the reduced model: 1 to 4% from placement alone.
- Holding the hazard fixed in gap space and widening the band moves the same exit inward.
- What signs the effect is where the hazard sits relative to the margin, not the width of the band.

Data: cleaning and price construction

- Non-promotion transactions only; unit-value prices in $\text{€}[0.20, 25]$, cent-rounded.
- Posted price: monthly mode per cell, kept with ≥ 20 transactions, modal share $\geq 50\%$.
- Lifecycles: span ≥ 12 months, density $\geq 1/2$, longest gap ≤ 6 months.
 - Left-censored excluded from entry, right-censored from exit.
- V-filter removes unflagged sales; change if $0.001 \leq |\Delta \log p| \leq \log(10/3)$.

Table: Descriptive statistics.

Complete panel		Value
Sample period	2008m1–2020m12	
Transactions (non-promotion)		163,524,586
Recorded purchase value (mn €)		341
Unique products (UPCs)		999,062
Brands		20,404
Retail chains		231
Product categories (groups)		719 (86)

Notes: The complete panel counts all non-promotion transactions, 2008–2020.

Whose price changes: product or chain?

- A chain posts one price nationwide: same price in every province 98% of the time.
- If a chain changes a product's price, another chain follows within a month 63%.
 - Placebo with shifted timing: 14%; matched changes agree in sign 94%.
- I read a price change as coming from the product, not the retailer.

► Table

► Back

Table: Geography of price setting.

	Rate	Observations
Same price in every province, within chain	0.98	51,907
Synchronization, ± 1 month	0.63	4,778
Same province only	0.62	799
Chain level	0.55	2,067
Same month only	0.46	4,778
Placebo: random time shifts	0.14	4,778
Same sign among matched changes	0.94	3,446

Notes: The first row covers the 51,907 product-chain-months of the pricing sample observed in two or more provinces, and reports the share in which the modal posted price is identical in every one of them; the mean standard deviation of the log price across provinces is 0.001, against 0.028 across chains in the same product-month (4,242 product-months). The remaining rows use the 185 products of the pricing sample sold in two or more chains (22,165 cell-months with a defined change indicator). An opportunity is a pair (product, month) in which one chain changes the product's price and a different chain sells the product with a defined change indicator in that month; a match occurs if the second chain changes the price within one month. Same province restricts the pair to cells in the same province. Chain level aggregates cells to the product-chain-month before pairing. The placebo shifts each cell's change months by a random offset of between three and 24 months (20 draws, mean reported). The last row is the share of matched cross-chain change pairs in which the two changes have the same sign.

What barcode exit measures

Table: What barcode exit measures.

	Share
National product exit within 6 months, revenue-weighted	0.82
products sold in 2+ chains	0.70
Same-brand entrant in same category, $[-3, +6]$ months	0.62
at a random month of the same brand-category (placebo)	0.50
excess over placebo	0.12
closest entrant: same / smaller / larger pack	0.34 / 0.30 / 0.34

Notes: Sample: valid lifecycles of the turnover sample (Table 1), interior months; 96,915 cell exits, of which 27,263 national product exits with an identified manufacturer brand. A national product exit occurs when the UPC leaves every chain and province in the panel within six months of the cell exit; shares are weighted by lifecycle revenue. The replacement rows report the share of national product exits followed by a new barcode of the same brand in the same category within three months before to six months after the exit; the placebo evaluates the same window at a random interior month for the same brand-category, averaged over ten draws. Pack size compares volume per unit of the exiting product and its closest replacement, when both share a measurement unit.

- Excess same-brand replacement tied to the exit itself: twelve percentage points.

Pricing does not change on the way out

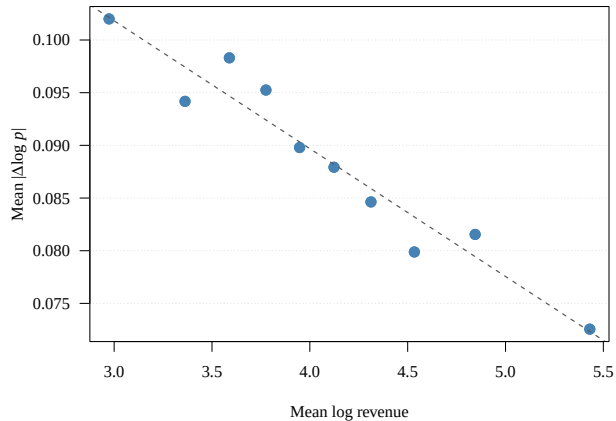
Table: Adjustment frequency by months before exit.

Months before exit	Frequency	<i>N</i>
1–3	0.079	5,825
4–6	0.100	4,896
7–12	0.088	10,823
13 or more	0.079	76,701
Censored (no observed exit)	0.079	70,834

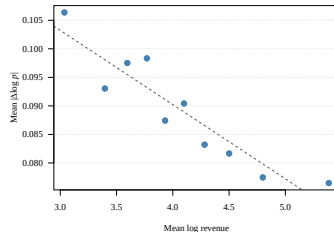
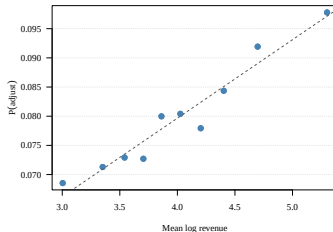
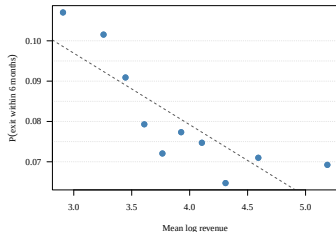
Notes: Mean monthly adjustment probability on the incumbent panel of the pricing sample, by months between the observation and the product's exit from the sample. The censored row collects products whose exit is never observed.

- What distinguishes exiting products is their level of revenue, not their timing.

Size of price changes by revenue decile



Robustness: trailing twelve-month size proxy



- Also robust to: national and brand exits, sample halves, quantity margin, within-brand.

► Back

Persistence of idiosyncratic productivity

- Monthly prices carry measurement error.
 - x_{it} : a log unit value, net of category median, product, month, and market share FE.
 - Desired prices are a markup over W_t/z_{it} , so x inherits the persistence of $\log z$,
 - and the noise: x carries the same error.
- Measurement error biases the variance of x and not its autocovariances:

$$x_{it} = x_{it}^* + u_{it} : \quad C_0 = \text{Var}(x^*) + \text{Var}(u), \quad C_k = \rho^k \text{Var}(x^*) \quad (k \geq 1)$$

- Ratios of successive autocovariances then recover ρ whatever the size of the error:

$$C_2/C_1 = 0.830, \quad C_3/C_2 = 0.833$$

Product-month cells, 3,550,549 consecutive pairs. Identification needs u uncorrelated over time within a product, not small.

Incumbent's problem

- Value, conditional on surviving this period's exit draw:

$$V(s) = \max\{V_N(s), V_A(s)\}$$

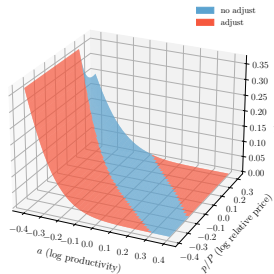
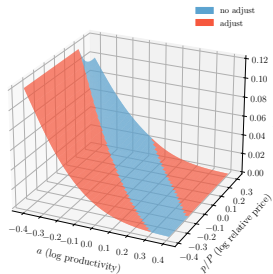
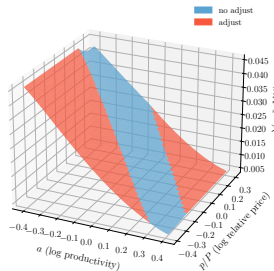
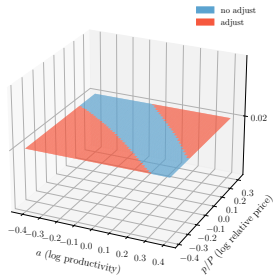
- Keep the inherited price, or pay the menu cost and reset:

$$\begin{aligned} V_N(p_{-1}, z, \mathcal{S}) &= \pi(p_{-1}, z, \mathcal{S}) + \mathbb{E}[\Xi(1 - \lambda(s')) V(p_{-1}, z', \mathcal{S}')] \\ V_A(z, \mathcal{S}) &= \max_p \left(\pi(p, z, \mathcal{S}) + \mathbb{E}[\Xi(1 - \lambda(s')) V(p, z', \mathcal{S}')] \right) \end{aligned}$$

- Flow profit carries the labor-denominated menu cost f when the price changes.

Ξ : stochastic discount factor. $\lambda(s')$: next-period exit hazard, so pricing lowers exit risk through V .

Exit hazard in two dimensions



Krusell-Smith solution algorithm

Linear rule $G(\mathcal{S}) = \bar{\pi} + b_1(\mathcal{S} - s^*)$ on a grid $\{\mathcal{S}_k\}_{k=1}^{50}$: s^* the state at which inflation equals trend.

- 1 Guess G_{old} with (s^*, b_1) .
- 2 VFI given $G \rightarrow$ price policy $p(s)$.
- 3 Solve for the stationary μ given p and G .
- 4 Push μ one period forward; deflating prices by G , the CES index implies

$$\hat{G}(\mathcal{S}_k) = G(\mathcal{S}_k) + \frac{1}{1-\theta} \log \mathbb{E}_{\hat{\mu}_k} \left[e^{(1-\theta)p(s-G(\mathcal{S}_k))} \right].$$

- 5 Fit $\hat{G}$ on $\mathcal{S}$ across grid points by WLS, weights $\propto$ ergodic mass; update (s^*, b_1) , get G_{new} .
- 6 Iterate to $\|G_{\text{new}} - G_{\text{old}}\|_{\infty} < \varepsilon$.

Unconditional variance of output

- Nominal demand follows the random walk with drift; draw $\eta_t \sim N(0, \sigma_\eta^2)$.
- Given the converged rule G , iterate forward

$$P_{t+1} = P_t + G(S_t - P_t), \quad Y_t = S_t - P_{t+1}$$

- Simulate 25,000 periods after burn-in and report the sample variance of Y_t .
- Selection raises $\text{Var}(Y)$ by about 15%; random churn alone moves it by 4%.

Backward-forward shooting

Outer iteration on the inflation path $\{\pi_t\}_{t=0}^{T-1}$, with $\mathcal{S}_t = S_t/P_{t-1}$:

- 1 Guess $\{\pi_t\}$; move the state forward from $\mathcal{S}_0 = \mathcal{S}^{ss} + \Delta$ by $\mathcal{S}_{t+1} = \mathcal{S}_t + \bar{\pi} - \pi_t$.
- 2 Backward pass: $V_T = V^{ss}$; one Bellman step at each $\mathcal{S}_t$, holding λ_t from the last iteration.
- 3 Forward pass: $\mu_0 = \mu^{ss}$; exit at λ_t , entry at replacement mass, reset, drift, deflation by π_t .
- 4 Realized inflation π_t^{new} : the CES index of the cross-section that has just repriced.
- 5 Update π_t until $\sup_t |\pi_t^{\text{new}} - \pi_t| < 10^{-7}$.

$\mathcal{S}^{ss}$ is re-solved by bisection so the stationary CES residual is exactly zero, which leaves the no-shock path flat. The reported response is $\psi_t = (\mathcal{S}_t - \mathcal{S}_t^0)/\Delta$, differenced against a second solve at $\Delta = 0$, over $T = 60$ months at $\Delta = 0.005$.

Cumulative response of output

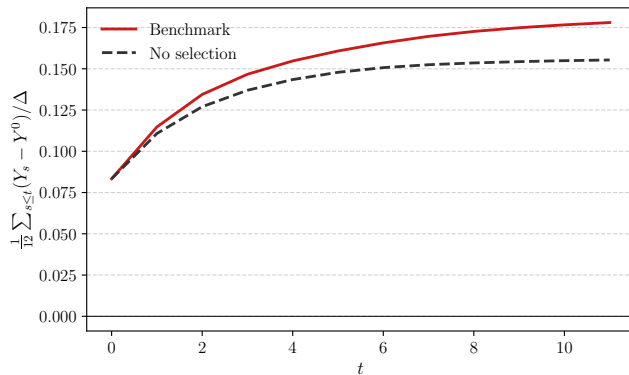
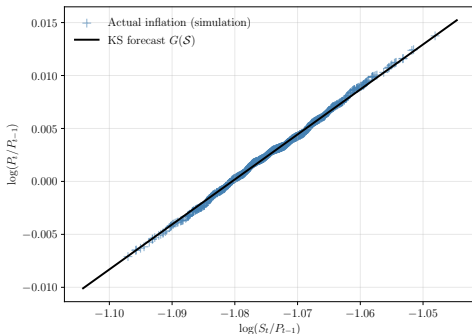
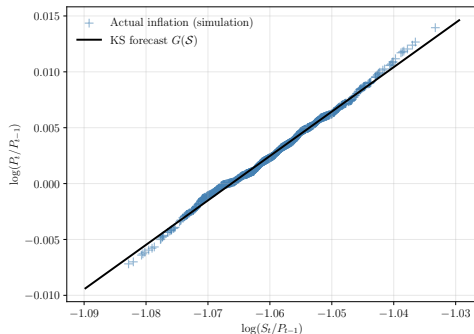


Table: Pricing moments and unconditional output variance.

Model	δ	ξ	τ	Frequency	Kurtosis	$\text{Var}(Y) \times 10^5$
Benchmark	0.02	20	5	0.086	1.41	2.36
No selection	0.02	0	0	0.088	1.32	2.05
No churn	0.00	0	0	0.092	1.31	1.97

- Kurt/freq: sufficient statistic for small nominal shocks (Alvarez et al., 2016).

Forecast errors: linear rule

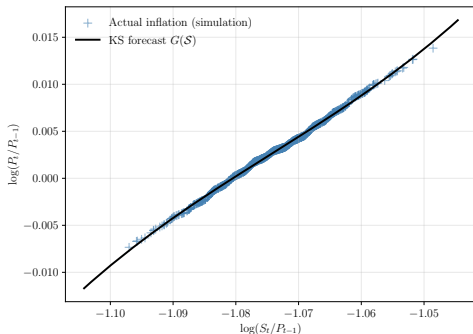
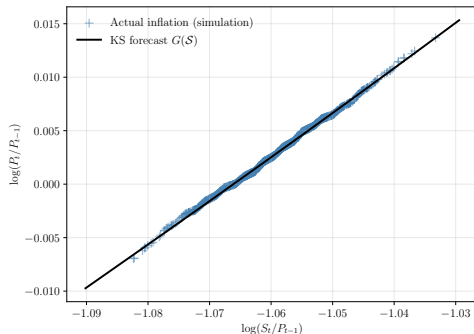


- Benchmark (left) bows at the support boundaries; No selection (right) is linear.

► Cubic rule

► Back

Forecast errors: cubic rule



- The bow disappears; the forecast-error dispersion drops by a third in the Benchmark.

► Moments

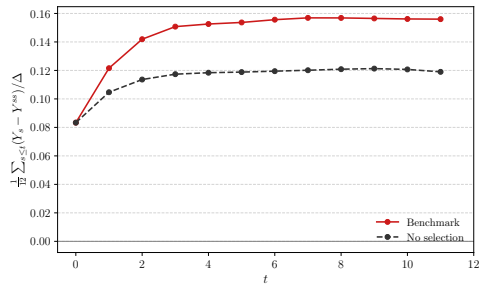
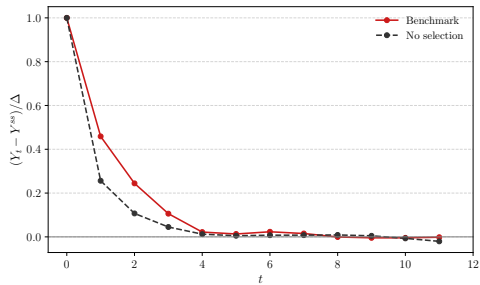
► Back

Table: Stationary moments under linear vs. cubic forecasting rule.

Model	Linear rule			Cubic rule		
	Frequency	Kurtosis	$\text{Var}(Y) \times 10^5$	Frequency	Kurtosis	$\text{Var}(Y) \times 10^5$
Benchmark	0.086	1.41	2.362	0.086	1.39	2.190
No selection	0.088	1.32	2.051	0.088	1.32	1.988
No churn	0.092	1.31	1.967	0.092	1.31	1.988

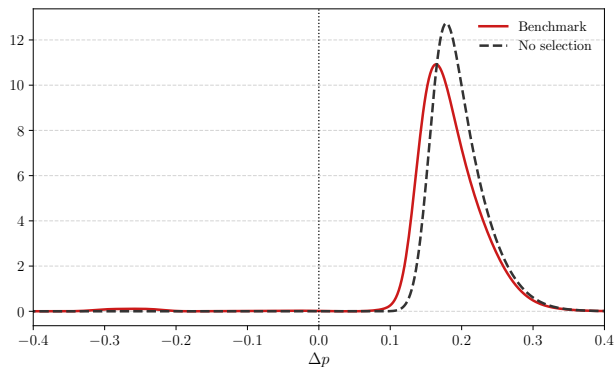
- Pricing moments unchanged; the variance gap shrinks from +15.4 to +10.3%.
- One-year CIRs differ by less than one percent across rules.

IRF and CIR under the cubic rule



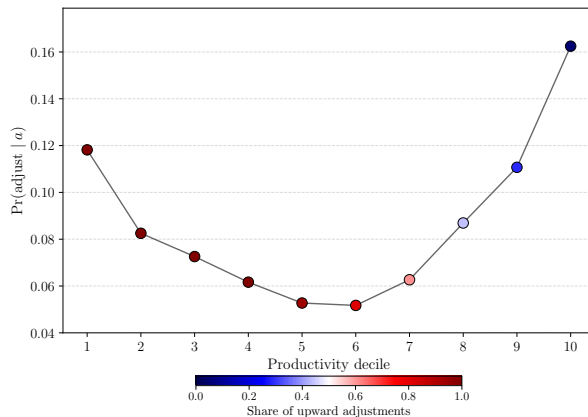
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Missing mass: lowest productivity quartile



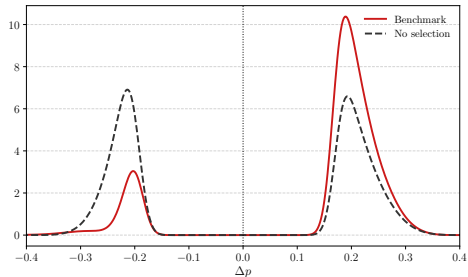
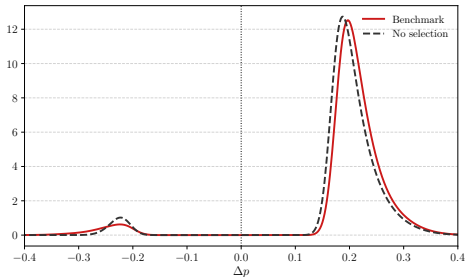
- The mass selection removes from Q1 is the upward resets exit intercepts.

Who adjusts, by productivity



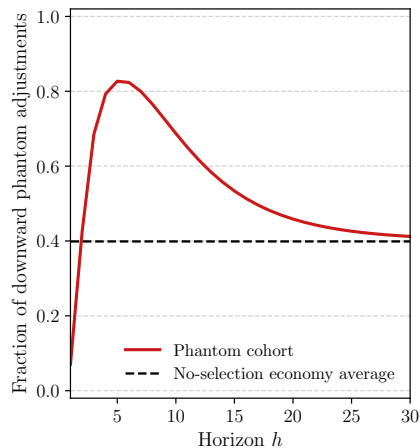
- Adjustment concentrates in the tails; selective exit removes the low- z , high-adjustment tail.

Missing mass: middle quartiles



- Less mass at moderate downward adjustments: the mean-reversion leg goes missing too.

Phantom cohort vs. uniform cohort



- The selectively removed cohort adjusts far more than a uniformly drawn one.

Where the interest rate is

- Log utility and $C = S/P$ give $1 = \beta(1 + i) \mathbb{E}[S/S']$: the nominal rate is constant.
- Policy is a constant growth target for nominal spending; a shock shifts its level for good.
- The Euler equation maps a rate path into a spending path: 25bp for a year shifts it 0.25%.

Response to a 0.25% permanent shift	Benchmark	No selection
Output on impact	+0.25%	+0.25%
Output, average over year one	+0.045%	+0.039%
Cumulated output, percent-months	0.53	0.47
Price level caught up after one month	62%	67%
Real rate, month one, annualized	-1.9pp	-2.0pp
Equivalent Calvo price duration	3.2 months	2.9 months

Computed from the transition paths behind the headline CIR. The equivalent duration solves

$\frac{1}{12} \sum_{s=0}^{11} (1 - \alpha)^{s+1} = \text{CIR}$. The mapping assumes the level shift is never reversed: a cut the bank later undoes leaves prices nearly unchanged and output tracking nominal spending.

Decomposition counterfactuals

- The hazard does two jobs: it discounts continuation values, and it deletes mass.
- I separate them, holding the flow δ fixed:
 - firms price under a perceived hazard λ_B in the Bellman equation,
 - the cross-section evolves under a realized hazard λ_F in the law of motion:

$$\lambda_j(s) = \delta \cdot \frac{\exp(-\xi_j V(s))}{\mathbb{E}_\mu[\exp(-\xi_j V)]}, \quad j \in \{B, F\}$$

- Each configuration (ξ_B, ξ_F, τ) is solved and shocked; one channel at a time.
 - Option value: $(20, 0)$; censoring: $(0, 20)$; interaction: residual to the full $(20, 20)$.

Exit carries the amplification; entry works against it

Economy	ξ	τ	CIR (1 year)	vs. No selection
No selection	0	0	0.155	—
Entry only	0	5	0.157	+0.8%
Exit only	20	0	0.184	+18.5%
Benchmark	20	5	0.178	+14.6%

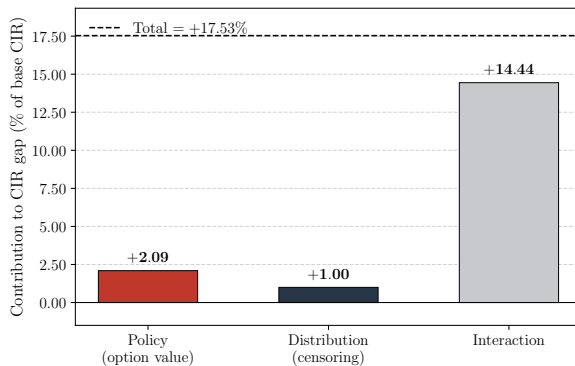
- Amplification needs the exiting firm to differ from the average firm in how soon it reprices.
- Selective entry puts freshly priced firms at low productivity, where the hazard would concentrate.
- But exit then lands on firms whose correction was not imminent.

[► Where exit lands](#)[► Decomposition](#)[► Back](#)

Where exit lands, with and without selective entry

Price gap x	Share of the exit flow		Share of the cross-section	
	Exit only	Benchmark	Exit only	Benchmark
$x < -0.25$	0.315	0.304	0.013	0.012
$-0.25 \leq x < -0.20$	0.284	0.258	0.023	0.022
$-0.20 \leq x < -0.15$	0.250	0.228	0.052	0.049
$-0.15 \leq x < -0.10$	0.103	0.078	0.098	0.095
$-0.10 \leq x < -0.05$	0.014	0.023	0.165	0.162
$-0.05 \leq x < 0$	0.013	0.044	0.206	0.202
$x \geq 0$	0.020	0.065	0.443	0.458
Mean price gap	-0.215	-0.198	-0.014	-0.010
Mean log productivity	-0.296	-0.316	0.032	0.021

CIR decomposition: entry-only to exit-only



- The standalone bars nearly vanish: neither job works without the other.

Moment decomposition: menu-cost model

Table: Decomposition of selection effects, menu-cost model.

Moment	M_0	M_1	$M_1 - M_0$	Δ_{policy}	Δ_{dist}	Δ_{int}
<i>Pair 1 — No-selection ($\xi = 0, \tau = 0$) vs Benchmark ($\xi = 20, \tau = 5$)</i>						
mean $\Delta \log p$	+0.032	+0.042	+0.010	-0.008	+0.015	+0.003
kurtosis $\Delta \log p$	1.318	1.415	+0.097	-0.022	+0.124	-0.006
frequency (pp)	8.785	8.581	-0.204	+0.050	-0.193	-0.061
<i>Pair 2 — entry-only ($\xi = 0, \tau = 5$) vs exit-only ($\xi = 20, \tau = 0$)</i>						
mean $\Delta \log p$	+0.012	+0.052	+0.040	-0.030	+0.047	+0.023
kurtosis $\Delta \log p$	1.203	1.508	+0.305	+0.016	+0.366	-0.076
frequency (pp)	8.746	8.496	-0.250	-0.963	-0.140	+0.853

Moment decomposition: Calvo model

Table: Decomposition of selection effects, Calvo model.

Moment	M_0	M_1	$M_1 - M_0$	Δ_{policy}	Δ_{dist}	Δ_{int}
<i>Pair 1 — No-selection ($\xi = 0, \tau = 0$) vs Benchmark ($\xi = 20, \tau = 5$)</i>						
mean $\Delta \log p$	+0.026	+0.026	-0.001	+0.059	-0.054	-0.006
skewness $\Delta \log p$	+0.377	+0.205	-0.173	+0.450	-0.580	-0.044
kurtosis $\Delta \log p$	3.667	4.795	+1.128	+0.612	+0.384	+0.132
<i>Pair 2 — entry-only ($\xi = 0, \tau = 5$) vs exit-only ($\xi = 20, \tau = 0$)</i>						
mean $\Delta \log p$	+0.018	+0.035	+0.016	+0.026	-0.003	-0.007
skewness $\Delta \log p$	+0.342	+0.701	+0.359	+0.354	+0.161	-0.156
kurtosis $\Delta \log p$	3.692	4.626	+0.934	+0.825	+0.092	+0.018

- No menu cost in flow profits; adjustment arrives with probability α .
- Values:

$$V_N(p, z, \mathcal{S}) = \pi(p, z, \mathcal{S}) + \mathbb{E}[\Xi(1 - \lambda(s'))((1 - \alpha)V_N(p, z', \mathcal{S}') + \alpha V_A(z', \mathcal{S}'))]$$
$$V_A(z, \mathcal{S}) = \max_{p'} V_N(p', z, \mathcal{S})$$

- Household, productivity, exit hazard, and entry block unchanged.
- In the timing, the pricing decision becomes an exogenous Bernoulli draw.

Table: Calvo model: pricing moments under selective exit, baseline calibration.

	ξ	τ	Frequency	$\sigma(\Delta \log p)$	skewness	kurtosis
No selection	0	0	0.088	0.081	+0.377	3.667
Benchmark	20	5	0.088	0.084	+0.205	4.795

- Frequency identical by construction; the action is in the shape of $\Delta \log p$.

Sensitivity to persistence: Calvo

Table: Calvo model: pricing moments by persistence and selection.

Parameterisation	ξ	τ	Frequency	$\sigma(\Delta \log p)$	$\text{skew}(\Delta \log p)$	Kurtosis
$\rho = 0.85, \sigma_\varepsilon = 0.086$	0	0	0.09	0.08	+0.38	3.67
	20	5	0.09	0.08	+0.20	4.79
$\rho = 0.95, \sigma_\varepsilon = 0.051$	0	0	0.09	0.09	+0.52	4.46
	20	5	0.09	0.08	+0.43	4.28

- At $\rho = 0.95$ the Calvo kurtosis ranking reverses: the policy channel needs fast reversion.

► Menu cost

► Back

Table: Menu-cost model: pricing moments by persistence and selection.

Parameterisation	ξ	τ	Frequency	$\sigma(\Delta \log p)$	$\text{skew}(\Delta \log p)$	Kurtosis
$\rho = 0.85, \sigma_\varepsilon = 0.086$	0	0	0.09	0.21	-0.41	1.32
	20	5	0.09	0.21	-0.45	1.41
$\rho = 0.95, \sigma_\varepsilon = 0.051$	0	0	0.06	0.17	-0.53	1.39
	20	5	0.06	0.16	-0.64	1.56

- The menu-cost kurtosis gap survives and slightly widens: +0.10 to +0.17.